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Time and Work(Pipes and Cistern)

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Question 1

A cistern is fitted with three taps, namely p, q, and r. P and Q can fill a cistern in 10 and 15 minutes respectively whereas R (emptying tap fitted at the bottom of the cistern) can empty it in 12 minutes. If all the three pipes are kept open, in how much time will the cistern be filled?

A) 12

B) 10

C) 7

D) 15

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Question 2

A cistern generally takes 20 minutes to be filled by a pipe, but due to a leak, it takes 10 extra minutes to get filled. Find the time in which the leak alone can empty the cistern filled with water.

A) 20

B) 30

C) 60

D) 35

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Question 3

Two taps X and Y can fill a cistern in 32 and 40 minutes respectively. Both the taps are opened into the empty cistern and after sometime tap X is closed. Tap Y alone fills the remaining portion of the cistern. If it took 25 minutes to fill the tank, for how much time (min) was tap X kept open?

A) 15

B) 16

C) 14

D) 12

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Question 4

Tap X and Y can fill a tank in 30 and 40 minutes respectively. Tap Z can empty the filled tank in 60 minutes. If all the three taps are kept open for one minute each, how much time (min) will the taps take to fill the tank?

A) 48

B) 72

C) 24

D) None of these

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Question 5

A large tanker can be filled by two pipes P and Q in 60 minutes and 40 minutes respectively. How many minutes will it take to fill the tanker from empty state, if Q is used for half the time and P and Q fill it together for other half?

A) 15 min

B) 20 min

C) 27.5 min

D) 30 min

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Question 6

Pipes A, B and C can fill a tank in 30, 60 and 120 minutes respectively. Pipes B and C are kept open for 10 minutes, and then Pipe B is shut while Pipe A is opened. Pipe C is closed 10 minutes before the tank overflows. How long does it take to fill the tank?

A) 40 minutes

B) 28 minutes

C) 30 minutes

D) 36 minutes

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Question 7

Bibhor takes 3 hours to fetch as much water as Ahmed can fetch in 2 hours. Deepak takes 5 hours to fetch as much water as Bibhor can fetch in 4 hours. A tank takes 20 hours to fill if all work together. What time would Bibhor take to fill the tank alone?

A) 50 hr

B) 56 hr

C) 77 hr

D) 66 hr

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Question 8

A tank is fitted with 8 pipes, some of which that fill the tank and others that empty the tank. Each of the pipes that fills the tank fills it in 8 hours, while each of those that empty the tank empties it in 6 hours. If all the pipes are kept open when the tank is full, it will take 6 hours to drain the tank. How many of these are fill pipes?

A) 2 fill pipes

B) 4 fill pipes

C) 6 fill pipes

D) 5 fill pipes

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Question 9

Pipe A, B and C are kept open and together fill a tank in t minutes. Pipe A is kept open throughout, pipe B is kept open for the first 10 minutes and then closed. Two minutes after pipe B is closed, pipe C is opened and is kept open till the tank is full. Each pipe fills an equal share of the tank. Furthermore, it is known that if pipe A and B are kept open continuously, the tank would be filled completely in t minutes. How long will C alone take to fill the tank?

A) 36

B) 27

C) 18

D) 24

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Question 10

A drain pipe can drain a tank in 12 hours, and a fill pipe can fill the same tank in 6 hours. A total of n pipes which include a few fill pipes and the remaining drain pipes can fill the entire tank in 2 hours. How many of the following values could ‘n’ take?

a) 24 b) 16 c) 33 d) 13 e) 9 f) 8

A) 3

B) 4

C) 2

D) 1

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Question 11

A fill pipe can fill a tank in 20 hours, a drain pipe can drain a tank in 30 hours. If a system of n pipes (fill pipes and drain pipes put together) can fill the tank in exactly 5 hours, which of the following are possible values of n (More than one option could be correct)?
1)32 2)54 3)29 4)40

A) 1 and 2 only

B) 1 and 3 only

C) 2 and 4 only

D) 2 and 3 only

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Question 12

A cistern of capacity 40 litres has an inlet and an outlet pipe. When both the pipes are opened at once, it takes 8 minutes to fill the cistern. However, if the outflow rate is increased 1.5 times, the cistern never gets filled. Which of the following can be the outflow rate?

A) 8 litres/minute

B) 6 litres/minute

C) 12 litres/minute

D) 9 litres/minute

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Question 13

A cistern of 475 litres is completely filled using pipes A and B, with Pipe A being open for 5 more hours than pipe B. If we are to interchange the operating hours of the two pipes than pipe A would have pumped half the water as much as pipe B, then find the time for which pipe B was open. Also, given that if the two pipes were open simultaneously the tank would fill in 19 hours.

A) 10 hr

B) 14 hr

C) 16 hr

D) 20 hr

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Question 14

If A and B are opened as inlet pipe into the cistern and C and D are opened as outlet pipes from the cistern and all the four pipes are opened simultaneously, how many hours will it take to fill the cistern completely?

A) 20 hours

B) 11(1/9) hours

C) 22 (2/9) hours

D) 45 hours

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Question 15

Two pipes A and B can fill a tank in 24 hours and 120/7 hours respectively. Harihar opened the pipes A and B to fill an empty tank and some times later he closed the taps A and B, when the tank was supposed to be full. After that it was found that the tank was emptied in 2.5 hours because an outlet pipe C' connected to the tank was open from the beginning. If Harihar closed the pipe C instead of closing pipes A and B the remaining tank would have been filled in :

A) 2 hours

B) 8 hours

C) 6 hours

D) 16 hours

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THANK YOU

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